

DATA-260 Homework 4

Name: Ayushi Shringi

SID4: 0036

DOMAIN_ID: 4

Domain Entity: Open-source package vulnerabilities

Port: 8036

Prefix: s0036

SEED: 0036

VERIFY_SEED: 260036

Python Version: Python 3.12.14

Embedding model: sentence-transformers/all-MiniLM-L6-v2

Generation model: google/flan-t5-base

Hardware: MacBook Pro

Chip: Apple M5 Pro

Memory: 24 GB

Operating System: macOS Tahoe 26.5.1

GitHub Repository: <https://github.com/ayushishringi/data260-0036/tree/hw4>

Code Commit Hash: f7ac59ba5cd6608cf34e61950b735c9b79025c02

Git Tag: hw4

Repo Struct-

```
tree -L 3 -I '.venv|node_modules|dist|__pycache__|.git|.DS
```

```
.
├── AGENT.md
├── DOMAIN_SCHEMA.md
├── Dockerfile
├── Makefile
├── README.md
├── agents_demo.py
├── agents_graph.py
├── app.js
├── backend
│   ├── __init__.py
│   ├── auth.py
│   ├── auth_api.py
│   ├── database.py
│   ├── main.py
│   ├── models.py
│   ├── records.json
│   ├── report_api.py
│   ├── schemas.py
│   └── security.py
├── corpus
│   ├── CORPUS_MANIFEST.json
│   ├── SOURCES.md
│   └── github_reviewed_advisories_snapshot.txt
├── deploy
│   └── ecs-task-definition.json
├── docker-compose.yml
├── frontend
│   ├── README.md
│   ├── index.html
│   ├── package-lock.json
│   ├── package.json
│   ├── public
│   │   ├── favicon.svg
│   │   └── icons.svg
│   ├── src
│   │   ├── App.css
│   │   ├── App.jsx
│   │   ├── CreateRecord.jsx
│   │   ├── DeleteRecord.jsx
│   │   ├── Home.jsx
│   │   ├── Login.jsx
│   │   ├── UpdateRecord.jsx
│   │   ├── assets
│   │   ├── index.css
│   │   └── main.jsx
│   └── vite.config.js
├── hw1_client.py
├── index.html
├── nginx.conf
├── outputs
│   └── hw3_retrieval_outputs.json
├── questions.yaml
├── rag_docs
│   ├── doc_01.txt
│   ├── doc_02.txt
│   ├── doc_03.txt
│   ├── doc_04.txt
│   └── doc_05.txt
├── reports
│   ├── hw01
│   │   ├── AI_USE.md
│   │   ├── METRICS.md
│   │   ├── RUN_LOG.txt
│   │   ├── cases
│   │   └── raw
```

```
doc_03.txt
reports
├── hw01
│   ├── AI_USE.md
│   ├── METRICS.md
│   ├── RUN_LOG.txt
│   ├── cases
│   ├── raw
│   ├── report.pdf
│   ├── screenshots
│   └── verification.json
├── hw02
│   ├── AI_USE.md
│   ├── METRICS.md
│   ├── RUN_LOG.txt
│   ├── cases
│   ├── raw
│   ├── report.pdf
│   └── verification.json
├── hw03
│   ├── AI_USE.md
│   ├── CORPUS_MANIFEST.json
│   ├── METRICS.md
│   ├── RUN_LOG.txt
│   ├── SOURCES.md
│   ├── questions.yaml
│   ├── raw
│   ├── report.pdf
│   └── verification.json
└── hw04
    ├── AI_USE.md
    ├── METRICS.md
    ├── RUN_LOG.txt
    ├── rag_corpus_manifest.json
    ├── rag_questions.yaml
    ├── raw
    ├── report.pdf
    └── verification.json
requirements.txt
scripts
├── add_hw4_index.py
├── build_corpus_manifest.py
├── create_tables.py
├── prepare_hw4_rag.py
├── run_hw02_experiments.py
├── run_hw3_retrieval.py
├── run_hw4_n1.py
├── run_hw4_rag.py
├── run_nondeterminism.py
├── seed_hw4_n1.py
├── verify_hw01.py
├── verify_hw02.py
├── verify_hw03.py
└── verify_hw4.py
src
├── __init__.py
├── model_client.py
└── retrieval.py
styles.css
templates
├── base.html
├── dashboard.html
├── home.html
└── login.html
```

MySQL table and Application Data

```
mysql> DESCRIBE users;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(120)	NO		NULL	
email	varchar(255)	NO	UNI	NULL	
password_hash	varchar(255)	NO		NULL	

```
4 rows in set (0.010 sec)
```

```
mysql> DESCRIBE sessions;
```

Field	Type	Null	Key	Default	Extra
id	varchar(128)	NO	PRI	NULL	
user_id	int	NO	MUL	NULL	
created_at	datetime	NO		NULL	
expires_at	datetime	NO		NULL	

```
4 rows in set (0.001 sec)
```

```
mysql> DESCRIBE vulnerability_reports;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
package_name	varchar(255)	NO	MUL	NULL	
affected_version	varchar(255)	NO		NULL	
submitter_email	varchar(255)	NO		NULL	
description	text	NO		NULL	
severity	varchar(20)	NO		NULL	
agreed_to_terms	tinyint(1)	NO		NULL	
submission_date	datetime	NO		NULL	

```
8 rows in set (0.001 sec)
```

```
mysql> DESCRIBE related_advisories;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
report_id	int	NO	MUL	NULL	
advisory_text	text	NO		NULL	

```
3 rows in set (0.001 sec)
```

database.py × schemas.py × main.py × Home.jsx × seed_hw4_n

backend > database.py > Base

```
1  import os
2  from collections.abc import Generator
3
4  from sqlalchemy import create_engine
5  from sqlalchemy.orm import DeclarativeBase, Session, sessionmaker
6
7
8  DATABASE_URL = os.getenv(
9      "DATABASE_URL",
10     "mysql+pymysql://root@127.0.0.1:3306/s0036_rel",
11 )
12
13
14 engine = create_engine(
15     DATABASE_URL,
16     pool_pre_ping=True,
17 )
18
19
20 class Base(DeclarativeBase):
21     pass
22
23
24 db_session_basedede26 = sessionmaker(
25     bind=engine,
26     autoflush=False,
27     autocommit=False,
28 )
29
30
31 def get_db() -> Generator[Session, None, None]:
32     db = db_session_basedede26()
33
34     try:
35         yield db
36     finally:
37         db.close()
```

```

8
9 class User(Base):
10     __tablename__ = "users"
11
12     id: Mapped[int] = mapped_column(
13         Integer,
14         primary_key=True,
15         autoincrement=True,
16     )
17
18     name: Mapped[str] = mapped_column(
19         String(120),
20         nullable=False,
21     )
22
23     email: Mapped[str] = mapped_column(
24         String(255),
25         unique=True,
26         nullable=False,
27     )
28
29     password_hash: Mapped[str] = mapped_column(
30         String(255),
31         nullable=False,
32     )
33
34     sessions: Mapped[list["LoginSession"]] = relationship(
35         back_populates="user",
36         cascade="all, delete-orphan",
37     )
38
39
40 class LoginSession(Base):
41     __tablename__ = "sessions"
42
43     id: Mapped[str] = mapped_column(
44         String(128),
45         primary_key=True,
46     )
47
48     user_id: Mapped[int] = mapped_column(
49         ForeignKey("users.id"),
50         nullable=False,
51     )
52
53     created_at: Mapped[datetime] = mapped_column(

```

backend > models.py > RelatedAdvisory

```
68 class VulnerabilityReport(Base):
69
70     )
71
72     affected_version: Mapped[str] = mapped_column(
73         String(255),
74         nullable=False,
75     )
76
77     submitter_email: Mapped[str] = mapped_column(
78         String(255),
79         nullable=False,
80     )
81
82     description: Mapped[str] = mapped_column(
83         Text,
84         nullable=False,
85     )
86
87     severity: Mapped[str] = mapped_column(
88         String(20),
89         nullable=False,
90     )
91
92     agreed_to_terms: Mapped[bool] = mapped_column(
93         Boolean,
94         nullable=False,
95     )
96
97     submission_date: Mapped[datetime] = mapped_column(
98         DateTime,
99         nullable=False,
100     )
101
102     advisories: Mapped[list["RelatedAdvisory"]] = relationship(
103         back_populates="report",
104         cascade="all, delete-orphan",
105     )
106
107
108 class RelatedAdvisory(Base):
109     __tablename__ = "related_advisories"
110
111     id: Mapped[int] = mapped_column(
112         Integer,
113         primary_key=True,
114         autoincrement=True,
115     )
116
117     report_id: Mapped[int] = mapped_column(
```



```

63         db.commit()
64
65         return session_token
66
67
68     def get_current_user(
69         session_token: str | None = Cookie(
70             default=None,
71             alias=SESSION_COOKIE_NAME,
72         ),
73         db: Session = Depends(get_db),
74     ) -> User:
75         if not session_token:
76             raise HTTPException(
77                 status_code=401,
78                 detail="Login required",
79             )
80
81         current_time = datetime.now(timezone.utc).replace(tzinfo=None)
82
83         database_session = db.scalar(
84             select(LoginSession).where(
85                 LoginSession.id == session_token,
86                 LoginSession.expires_at > current_time,
87             )
88         )
89
90         if database_session is None:
91             raise HTTPException(
92                 status_code=401,
93                 detail="Login required",
94             )
95
96         user = db.get(User, database_session.user_id)
97
98         if user is None:
99             raise HTTPException(
100                 status_code=401,
101                 detail="Login required",
102             )
103
104         return user

```

```

23 @router.post("/register", status_code=201)
24 def register(
25     payload: RegisterRequest,
26     db: Session = Depends(get_db),
27 ) -> dict:
28     existing_user = db.scalar(
29         select(User).where(User.email == payload.email)
30     )
31
32     if existing_user is not None:
33         raise HTTPException(
34             status_code=409,
35             detail="Email already registered",
36         )
37
38     user = User(
39         name=payload.name.strip(),
40         email=str(payload.email),
41         password_hash=hash_password(payload.password),
42     )
43
44     db.add(user)
45     db.commit()
46     db.refresh(user)
47
48     return {
49         "id": user.id,
50         "name": user.name,
51         "email": user.email,
52     }
53
54
55 @router.post("/login")
56 def login(
57     payload: LoginRequest,
58     response: Response,
59     db: Session = Depends(get_db),
60 ) -> dict:
61     user = db.scalar(
62         select(User).where(User.email == payload.email)
63     )
64
65     if user is None or not verify_password(
66         payload.password,
67         user.password_hash,
68     ):
69         raise HTTPException(
70             status_code=401

```

```

56     def login(
57         raise HTTPException(
58             status_code=401,
59             detail="Invalid email or password",
60         )
61     )
62
63     session_token = create_server_session(db, user)
64
65     response.set_cookie(
66         key=SESSION_COOKIE_NAME,
67         value=session_token,
68         httponly=True,
69         samesite="lax",
70         secure=False,
71         max_age=1800,
72     )
73
74     return {
75         "id": user.id,
76         "name": user.name,
77         "email": user.email,
78     }
79
80 @router.get("/me")
81 def get_me(
82     current_user: User = Depends(get_current_user),
83 ) -> dict:
84     return {
85         "id": current_user.id,
86         "name": current_user.name,
87         "email": current_user.email,
88     }
89
90
91
92 @router.post("/logout", status_code=204)
93 def logout(
94     response: Response,
95     session_token: str | None = Cookie(
96         default=None,
97         alias=SESSION_COOKIE_NAME,
98     ),
99     current_user: User = Depends(get_current_user),
100     db: Session = Depends(get_db),
101 ) -> None:
102     if session_token is not None:
103         database_session = db.get(
104             LoginSession

```

```

● ((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % mysql -u root -D s0036_rel -e "SHOW TABLES;"
+-----+
| Tables_in_s0036_rel |
+-----+
| related_advisories  |
| sessions            |
| users               |
| vulnerability_reports |
+-----+

```

MySQL table and stored application data

```

● ((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % mysql -u root -D s0036_rel -e "
SELECT COUNT(*) AS users FROM users;
SELECT COUNT(*) AS sessions FROM sessions;
SELECT COUNT(*) AS reports FROM vulnerability_reports;
"
+-----+
| users |
+-----+
|      1 |
+-----+
+-----+
| sessions |
+-----+
|        15 |
+-----+
+-----+
| reports |
+-----+
|       5000 |
+-----+

```

Login Page

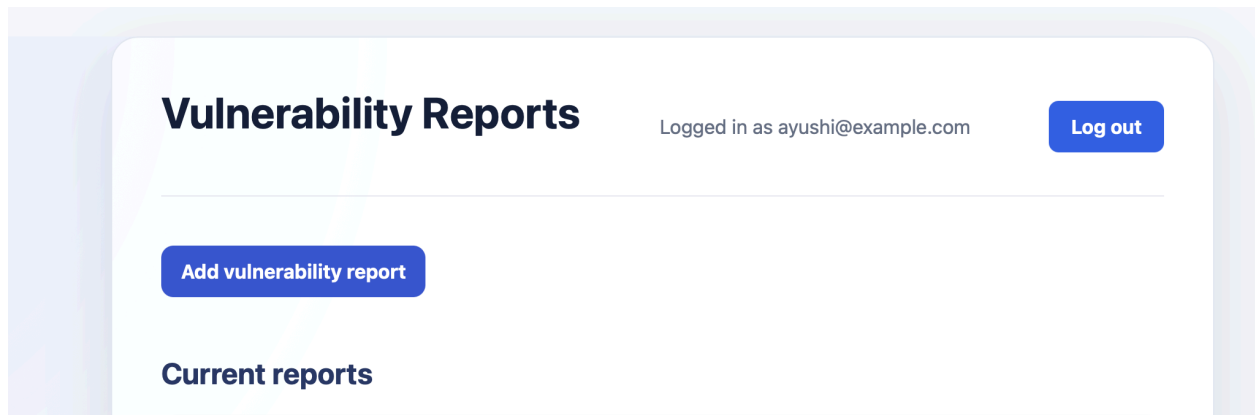
Login

Email

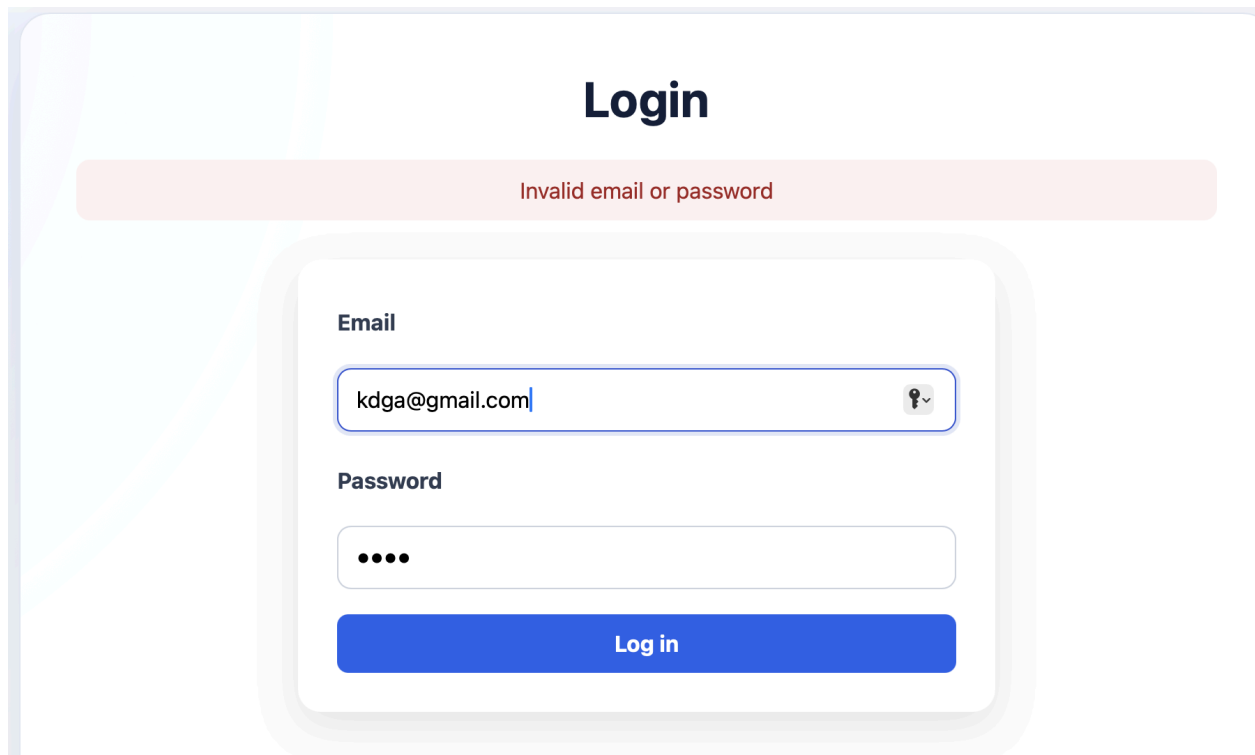
Password

[Log in](#)

Home Page



Invalid Login



React CRUD Interface

```

32     }
33   />
34
35   <Route
36     path="/"
37     element={
38       user ? (
39         <Home user={user} onLogout={handleLogout} />
40       ) : (
41         <Navigate to="/login" replace />
42       )
43     }
44   />
45
46   <Route
47     path="/create"
48     element={
49       user ? (
50         <CreateRecord />
51       ) : (
52         <Navigate to="/login" replace />
53       )
54     }
55   />
56
57   <Route
58     path="/update/:id"
59     element={
60       user ? (
61         <UpdateRecord />
62       ) : (
63         <Navigate to="/login" replace />
64       )
65     }
66   />
67
68   <Route
69     path="*"
70     element={<Navigate to="/" replace />}
71   />
72 </Routes>

```

```

28     }
29
30     useEffect(() => {
31         loadReports();
32     }, []);
33
34     async function handleLogout() {
35         await fetch(`${API_BASE}/api/auth/logout`, {
36             method: "POST",
37             credentials: "include",
38         });
39
40         onLogout();
41         navigate("/login");
42     }
43
44     async function handleDelete(reportId) {
45         const confirmed = window.confirm(
46             "Are you sure you want to delete this report?"
47         );
48
49         if (!confirmed) {
50             return;
51         }
52
53         try {
54             const response = await fetch(
55                 `${API_BASE}/api/reports/${reportId}`,
56                 {
57                     method: "DELETE",
58                     credentials: "include",
59                 }
60             );
61
62             if (!response.ok) {
63                 const data = await response.json().catch(() => ({}));
64
65                 throw new Error(
66                     data.detail || "Could not delete report."
67                 );
68             }
69
70             setReports((previousReports) =>
71                 previousReports.filter(
72                     (report) => report.id !== reportId
73                 )
74             );
75         } catch (err) {
76             setError(err.message);
77         }
78     }

```

```

44   async function handleDelete(reportId) {
76       |   setError(err.message);
77       | }
78   }
79
80   const visibleReports = reports.filter(
81       (report) =>
82       |   !report.packageName.startsWith("hw4-n1-package-")
83   );
84
85   return (
86       <main className="card">
87           <header>
88               <h1>Vulnerability Reports</h1>
89
90               <p>Logged in as {user.email}</p>
91
92               <button
93                   |   type="button"
94                   |   onClick={handleLogout}
95               >
96                   |   Log out
97               </button>
98           </header>
99
100          <hr />
101
102          <Link className="add-report-button" to="/create">
103              Add vulnerability report
104          </Link>
105
106          <h2>Current reports</h2>
107
108          {error && <p className="error">{error}</p>}
109
110          {visibleReports.length === 0 && !error && (
111              <p>No vulnerability reports found.</p>
112          )}
113
114          {visibleReports.map((report) => (
115              <article className="report-card" key={report.id}>
116                  <h3>
117                      {report.packageName} – ID {report.id}
118                  </h3>
119

```

```

122     </p>
123
124     <p>
125         Severity: {report.severity}
126     </p>
127
128     <p>{report.description}</p>
129
130     <div className="report-actions">
131         <Link
132             className="action-button update-button"
133             to={` /update/${report.id}`}
134         >
135             Update
136         </Link>
137
138         <button
139             className="action-button delete-button"
140             type="button"
141             onClick={() => handleDelete(report.id)}
142         >
143             Delete
144         </button>
145     </div>
146 </article>
147     )}
148 </main>
149 );
150 }
151
152 export default Home;

```

```

15  function App() {
58      <Route
59          path="/update/:id"
60          element={
61              user ? (
62                  <UpdateRecord />
63              ) : (
64                  <Navigate to="/login" replace />
65              )
66          }
67      />
68
69      <Route
70          path="/delete/:id"
71          element={
72              user ? (
73                  <DeleteRecord />
74              ) : (
75                  <Navigate to="/login" replace />
76              )
77          }
78      />
79
80      <Route
81          path="*"
82          element={<Navigate to="/" replace />}
83      />
84  </Routes>
85  </BrowserRouter>
86  );
87  }
88
89  export default App;

```

frontend > src > DeleteRecord.jsx > default

```
6  function DeleteRecord() {
13    async function handleDelete() {
18      const response = await fetch(
19        `${API_BASE}/api/reports/${id}`,
20        {
21          method: "DELETE",
22          credentials: "include",
23        }
24      );
25
26      if (!response.ok) {
27        const data = await response.json().catch(() => ({}));
28
29        throw new Error(
30          data.detail || "Could not delete report."
31        );
32      }
33
34      navigate("/");
35    } catch (err) {
36      setError(err.message);
37      setDeleting(false);
38    }
39  }
40
41  return (
42    <main className="card form-card">
43      <h1>Delete Vulnerability Report</h1>
44
45      <p>
46        Are you sure you want to permanently delete report ID {id}?
47      </p>
48
49      {error && <p className="error">{error}</p>}
50
51      <div className="form-actions">
52        <button
53          type="button"
54          className="action-button delete-button"
55          onClick={handleDelete}
56          disabled={deleting}
57        />
```

Successful Server Side Login Api Response

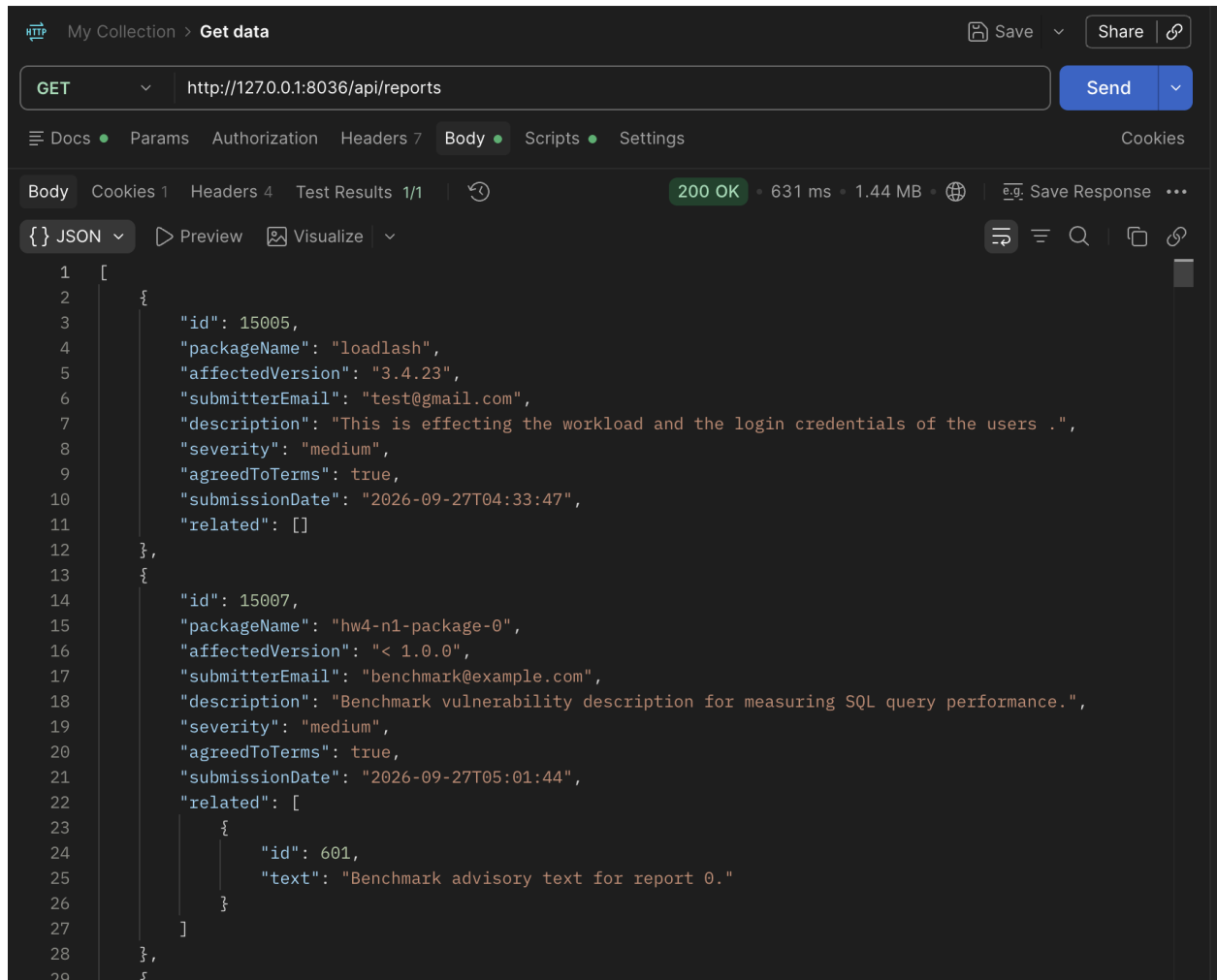
The screenshot displays a REST client interface with the following details:

- Method:** POST
- URL:** http://127.0.0.1:8036/api/auth/login
- Body (raw):**

```
1 {  
2   "email": "ayushi@example.com",  
3   "password": "data260"  
4 }
```
- Status:** 200 OK
- Response Time:** 78 ms
- Response Size:** 326 B
- Response Body (JSON):**

```
1 {  
2   "id": 1,  
3   "name": "Ayushi Shringi",  
4   "email": "ayushi@example.com"  
5 }
```

GET all reports API response



GET report by ID Api response

My Collection > Get data

GET Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

1

Body Cookies 1 Headers 4 Test Results 1/1 200 OK • 11 ms • 407 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "id": 15005,
3   "packageName": "loadlash",
4   "affectedVersion": "3.4.23",
5   "submitterEmail": "test@gmail.com",
6   "description": "This is effecting the workload and the login credentials of the users .",
7   "severity": "medium",
8   "agreedToTerms": true,
9   "submissionDate": "2026-09-27T04:33:47",
10  "related": []
11 }
```

POST create a new report Api response

My Collection > Get data

POST Send

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```
1 {
2   "packageName": "postman-test-package",
3   "affectedVersion": "< 1.0.1",
4   "submitterEmail": "ayushi@example.com",
5   "description": "This temporary report is created to test the API.",
6   "severity": "medium",
7   "agreedToTerms": true
8 }
```

Body Cookies 1 Headers 4 Test Results 0/1 201 Created • 33 ms • 407 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "id": 20008,
3   "packageName": "postman-test-package",
4   "affectedVersion": "< 1.0.1",
5   "submitterEmail": "ayushi@example.com",
6   "description": "This temporary report is created to test the API.",
7   "severity": "medium",
8   "agreedToTerms": true,
9   "submissionDate": "2026-09-27T22:28:40",
10  "related": []
11 }
```

PUT Update report API response

The screenshot shows a Postman interface for a PUT request. The request is sent to `http://127.0.0.1:8036/api/reports/20008`. The request body is raw JSON, containing details about a package update. The response is a 200 OK status with a JSON body that includes an ID, package name, affected version, submitter email, description, severity, agreed-to terms, submission date, and related reports.

Request:

```
1 {
2   "packageName": "postman-test-package-updated",
3   "affectedVersion": "< 1.0.2",
4   "submitterEmail": "ayushi@example.com",
5   "description": "This temporary report has been updated through the API.",
6   "severity": "high",
7   "agreedToTerms": true
8 }
```

Response:

```
1 {
2   "id": 20008,
3   "packageName": "postman-test-package-updated",
4   "affectedVersion": "< 1.0.2",
5   "submitterEmail": "ayushi@example.com",
6   "description": "This temporary report has been updated through the API.",
7   "severity": "high",
8   "agreedToTerms": true,
9   "submissionDate": "2026-09-27T22:28:40",
10  "related": []
11 }
```

DELETE report Api response

The screenshot shows a Postman interface for a DELETE request. The request is sent to `http://127.0.0.1:8036/api/reports/15005`. The response is a 204 No Content status, indicating that the report has been successfully deleted.

Request:

```
1
2
```

Response:

```
1
```

Home Page showing existing reports

Vulnerability Reports

Logged in as ayushi@example.com

Log out

Add vulnerability report

Current reports

loadlash — ID 15005

Affected version: 3.4.23

Severity: high

This is effecting the workload and the login credentials of the users .

Update

Delete

Create new vulnerability report form

Add Vulnerability Report

[← Back to reports](#)

Package name

TEST- 1234

Affected version

2.31.4

Submitter email

test@gmail.com

Description

This is s test report which im filling

Severity

Low



I agree to the terms.

Submit report

New Report Added in the dashboard report list

Vulnerability Reports

Logged in as ayushi@example.com

Log out

Add vulnerability report

Current reports

loadlash — ID 15005

Affected version: 3.4.23

Severity: high

This is effecting the workload and the login credentials of the users .

Update

Delete

TEST- 1234 — ID 15006

Affected version: 2.31.4

Severity: low

This is s test report which im filling

Update

Delete

Updating existing report

Update Vulnerability Report

[← Back to reports](#)

Package name

loadlash

Affected version

3.4.23

Submitter email

test@gmail.com

Description

This is effecting the workload and the login credentials of the users .

Severity

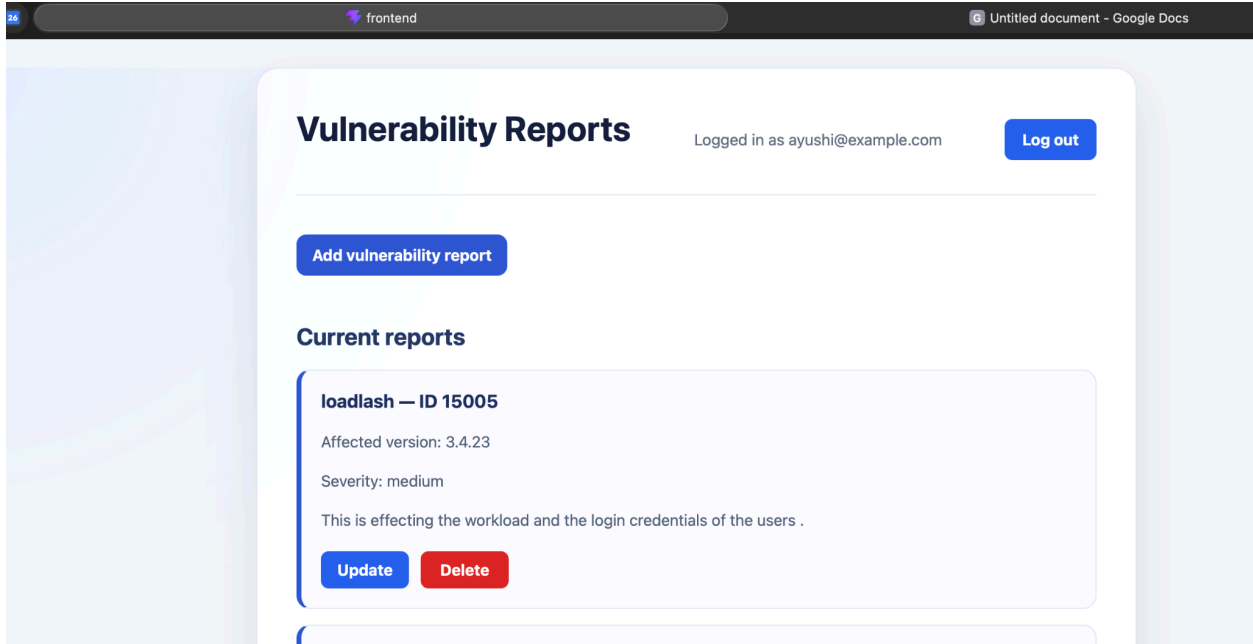
High



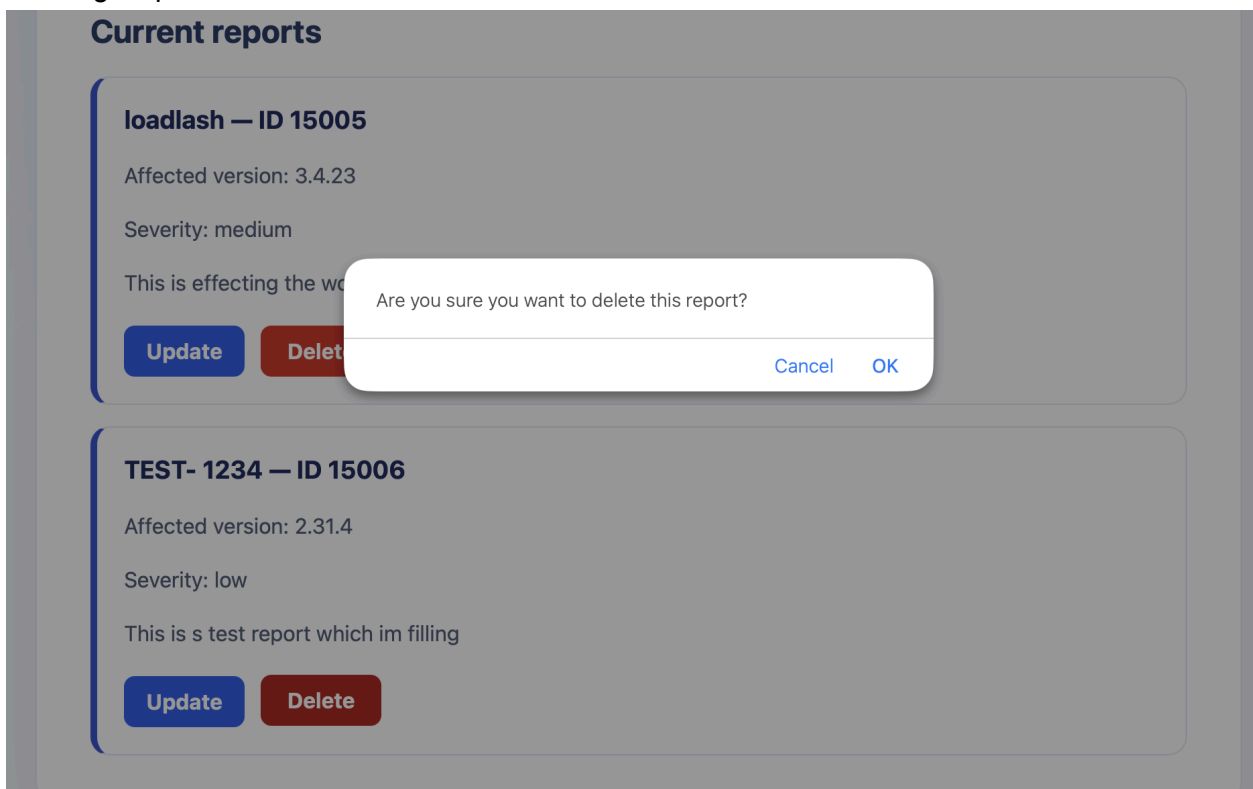
I agree to the terms.

Save changes

Updated Report (changed severity)



Deleting Report



Report Deleted

[Add vulnerability report](#)

Current reports

loadlash — ID 15005

Affected version: 3.4.23

Severity: medium

This is effecting the workload and the login credentials of the users .

[Update](#)

[Delete](#)

N+1 Query Problem and Optimization

```

18
19 def report_to_dict(report: VulnerabilityReport) -> dict:
20     return {
21         "id": report.id,
22         "packageName": report.package_name,
23         "affectedVersion": report.affected_version,
24         "submitterEmail": report.submitter_email,
25         "description": report.description,
26         "severity": report.severity,
27         "agreedToTerms": report.agreed_to_terms,
28         "submissionDate": report.submission_date.isoformat(),
29         "related": [
30             {
31                 "id": advisory.id,
32                 "text": advisory.advisory_text,
33             }
34             for advisory in report.advisories
35         ],
36     }
37
38
39 @router.get("")
40 def list_reports(
41     db: Session = Depends(get_db),
42     current_user: User = Depends(get_current_user),
43 ) -> list[dict]:
44     reports = db.scalars(
45         select(VulnerabilityReport)
46         .options(
47             selectinload(VulnerabilityReport.advisories)
48         )
49         .order_by(VulnerabilityReport.id)
50     ).all()
51
52
53     return [report_to_dict(report) for report in reports]
54

```

```

10 REPORT_COUNT = 5000
11 ADVISORY_COUNT = 200
12
13
14 def main() -> None:
15     db = db_session_basede26()
16
17     try:
18         # Remove only older benchmark rows created by this script.
19         old_reports = db.scalars(
20             select(VulnerabilityReport).where(
21                 VulnerabilityReport.package_name.like(
22                     f"{BENCHMARK_PREFIX}%"
23                 )
24             )
25         ).all()
26
27         for report in old_reports:
28             db.delete(report)
29
30         db.commit()
31
32         # Create 5,000 main vulnerability reports.
33         reports = []
34
35         for index in range(REPORT_COUNT):
36             reports.append(
37                 VulnerabilityReport(
38                     package_name=f"{BENCHMARK_PREFIX}{index}",
39                     affected_version="< 1.0.0",
40                     submitter_email="benchmark@example.com",
41                     description=(
42                         "Benchmark vulnerability description for "
43                         "measuring SQL query performance."
44                     ),
45                     severity="medium",
46                     agreed_to_terms=True,
47                     submission_date=datetime.now(
48                         timezone.utc
49                     ).replace(tzinfo=None),
50                 )
51             )
52
53         db.add_all(reports)
54         db.flush()
55
56         # Create 200 related advisories for the first 200 reports.
57         advisories = []
58
59         for index in range(ADVISORY_COUNT):
60             advisories.append(

```

seed_hw4_n1.py × run_hw4_n1.py × App.jsx × # index.css × UpdateReco

scripts > run_hw4_n1.py > ...

```
24 def run_naive(page_size: int) -> tuple[float, int, int]:
25     """
26
27     db = db_session_basede26()
28     query_counter = {"count": 0}
29
30     def count_query(*args, **kwargs):
31         query_counter["count"] += 1
32
33     event.listen(engine, "before_cursor_execute", count_query)
34
35     try:
36         start = time.perf_counter()
37
38         reports = db.scalars(
39             select(VulnerabilityReport)
40             .where(
41                 VulnerabilityReport.package_name.like(
42                     "hw4-n1-package-%"
43                 )
44             )
45             .order_by(VulnerabilityReport.id)
46             .limit(page_size)
47         ).all()
48
49         advisory_count = sum(
50             len(report.advisories)
51             for report in reports
52         )
53
54         elapsed_ms = (time.perf_counter() - start) * 1000
55
56         return (
57             elapsed_ms,
58             query_counter["count"],
59             advisory_count,
60         )
61
62     finally:
63         event.remove(engine, "before_cursor_execute", count_query)
64         db.close()
65
66 def run_fixed(page_size: int) -> tuple[float, int, int]:
67     """
68     Fixed version:
69     selectinload fetches all related advisories in one additional query,
70     instead of one query per report.
71     """
72
73
```

```

91         options\
92         selectinload(
93             VulnerabilityReport.advisories
94         )
95     )
96     .where(
97         VulnerabilityReport.package_name.like(
98             "hw4-n1-package-%"
99         )
100     )
101     .order_by(VulnerabilityReport.id)
102     .limit(page_size)
103 ).all()
104
105 advisory_count = sum(
106     len(report.advisories)
107     for report in reports
108 )
109
110 elapsed_ms = (time.perf_counter() - start) * 1000
111
112 return (
113     elapsed_ms,
114     query_counter["count"],
115     advisory_count,
116 )
117
118 finally:
119     event.remove(engine, "before_cursor_execute", count_query)
120     db.close()
121
122
123 def percentile(values: list[float], percentage: float) -> float:
124     sorted_values = sorted(values)
125     position = int(round((percentage / 100) * len(sorted_values))) - 1
126     position = max(0, min(position, len(sorted_values) - 1))
127     return sorted_values[position]
128
129
130 def summarize(values: list[float]) -> dict:
131     return {
132         "count": len(values),
133         "mean_ms": round(statistics.mean(values), 4),
134         "p50_ms": round(percentile(values, 50), 4),
135         "p95_ms": round(percentile(values, 95), 4),
136         "p99_ms": round(percentile(values, 99), 4),
137         "min_ms": round(min(values), 4),
138         "max_ms": round(max(values), 4),
139     }
140
141
142 def main() -> None:
143     os.makedirs(OUTPUT_DIR, exist_ok=True)
144

```

```

142     def main() -> None:
152         "page_sizes": PAGE_SIZES,
153         "conditions": {},
154     }
155
156     for page_size in PAGE_SIZES:
157         for variant, function in [
158             ("naive", run_naive),
159             ("fixed", run_fixed),
160         ]:
161             for _ in range(WARMUPS):
162                 function(page_size)
163
164             latencies = []
165             query_counts = []
166             advisory_counts = []
167
168             for run_number in range(1, RUNS + 1):
169                 latency_ms, query_count, advisory_count = (
170                     function(page_size)
171                 )
172
173                 latencies.append(latency_ms)
174                 query_counts.append(query_count)
175                 advisory_counts.append(advisory_count)
176
177                 rows.append(
178                     {
179                         "variant": variant,
180                         "page_size": page_size,
181                         "run": run_number,
182                         "latency_ms": round(latency_ms, 4),
183                         "query_count": query_count,
184                         "advisory_count": advisory_count,
185                     }
186                 )
187
188             key = f"{variant}_page_{page_size}"
189
190             summary["conditions"][key] = {
191                 "variant": variant,
192                 "page_size": page_size,
193                 "latency": summarize(latencies),
194                 "query_count": {
195                     "min": min(query_counts),
196                     "max": max(query_counts),

```

```

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % PYTHONPATH=. .venv/bin/python3 scripts/seed_hw4_n1.py
Seeded 5000 vulnerability reports and 200 related advisories.
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

Explain output before adding the package name index

```

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % mysql -u root -D s0036_rel -e \
"EXPLAIN
SELECT id, package_name, severity
FROM vulnerability_reports
WHERE package_name = 'hw4-n1-package-0 ';"
+-----+
| EXPLAIN |
+-----+
| -> Filter: (vulnerability_reports.package_name = 'hw4-n1-package-0 ') (cost=517 rows=493)
| -> Table scan on vulnerability_reports (cost=517 rows=4931)
|
+-----+

```

Package-name index added to vulnerability report

```

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % mysql -u root -D s0036_rel -e "SHOW INDEX FROM vulnerability_reports;"
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Nu |
| Index_type | Comment | Index_comment | Visible | Expression | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| vulnerability_reports | 0 | PRIMARY | 1 | id | A | 4931 | NULL | NULL | Nu | |
| BTREE | | | YES | NULL | | | | | | |
| vulnerability_reports | 1 | idx_vulnerability_reports_package_name | 1 | package_name | A | 4931 | NULL | NULL | Nu |
| BTREE | | | YES | NULL | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

Explain output after adding package name index

```

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % mysql -u root -D s0036_rel -e "EXPLAIN SELECT id, package_name, severity FROM vulnerability_report
s WHERE package_name = 'hw4-n1-package-0';"
+-----+
| EXPLAIN |
+-----+
| -> Index lookup on vulnerability_reports using idx_vulnerability_reports_package_name (package_name = 'hw4-n1-package-0') (cost=0.35 rows=1)
|
+-----+
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

N+1 Benchmark Data Seeding

```

Seeded 5000 vulnerability reports and 200 related advisories.
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % PYTHONPATH=. .venv/bin/python3 scripts/run_hw4_n1.py
Wrote raw benchmark rows to reports/hw04/raw/n1_benchmark.csv
Wrote benchmark summary to reports/hw04/raw/n1_benchmark.json
naive_page_10 p50= 1.5876 p95= 2.0612 p99= 2.4728 queries= 11 - 11
fixed_page_10 p50= 0.499 p95= 0.5347 p99= 0.5422 queries= 2 - 2
naive_page_50 p50= 6.2952 p95= 6.5629 p99= 10.8229 queries= 51 - 51
fixed_page_50 p50= 0.9184 p95= 0.9642 p99= 0.983 queries= 2 - 2
naive_page_200 p50= 25.4942 p95= 28.6356 p99= 30.2458 queries= 201 - 201
fixed_page_200 p50= 2.3868 p95= 2.4571 p99= 6.8399 queries= 2 - 2
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

Naive versus fixed N+1 Benchmark results

Page size	Version	SQL statements	p50 latency	p95 latency	p99 latency
10	Naive	11	1.6666 ms	2.1878 ms	2.3425 ms

10	Fixed	2	0.5055 ms	0.5466 ms	0.5773 ms
50	Naive	51	6.4875 ms	7.3788 ms	11.0407 ms
50	Fixed	2	0.9624 ms	1.0918 ms	1.4295 ms
200	Naive	201	25.5755 ms	28.3768 ms	31.1851 ms
200	Fixed	2	2.4388 ms	2.5665 ms	7.1390 ms

The fixed implementation was faster at every page size. For page size 10, the fixed version was approximately 3.30x faster, with a 69.7% latency reduction. For page size 50, it was approximately 6.74x faster, with an 85.2% latency reduction. For page size 200, it was approximately 10.49x faster, with a 90.5% latency reduction. The naive implementation grows approximately linearly because it performs $N+1$ SQL queries: one query for the reports and one additional query for each report's related advisories. The optimized implementation uses selectinload and remains at two SQL queries regardless of page size.

Offline Retrieval Augmented Generation Experiment

```

11
12 ROOT = Path(__file__).resolve().parents[1]
13 DOCS_DIR = ROOT / "rag_docs"
14 QUESTIONS_FILE = ROOT / "reports" / "hw04" / "rag_questions.yaml"
15 OUTPUT_DIR = ROOT / "reports" / "hw04" / "raw"
16
17 MODEL_NAME = "sentence-transformers/all-MiniLM-L6-v2"
18 K_VALUES = [1, 2, 3, 5]
19 STRATEGIES = ["no_context", "basic_context", "engineered_context"]
20
21
22 def normalize(text: str) -> set[str]:
23     words = re.findall(r"[a-z0-9.]+", text.lower())
24     return set(words)
25
26
27 def load_documents() -> list[dict]:
28     documents = []
29
30     for path in sorted(DOCS_DIR.glob("doc_*.txt")):
31         text = path.read_text(encoding="utf-8")
32         json_start = text.index("{")
33         data = json.loads(text[json_start:])
34
35         documents.append(
36             {
37                 "document_id": path.stem,
38                 "text": text,
39                 "data": data,
40             }
41         )
42
43     return documents
44
45
46 def load_questions() -> list[dict]:
47     return yaml.safe_load(
48         QUESTIONS_FILE.read_text(encoding="utf-8")
49     )
50
51
52 def fixed_versions(data: dict) -> list[str]:
53     versions = []
54
55     for affected in data.get("affected", []):
56         for version_range in affected.get("ranges", []):
57             for event in version_range.get("events", []):
58                 if "fixed" in event:
59                     versions.append(event["fixed"])
60
61     return versions
62
63
64 def package_names(data: dict) -> list[str]:
65     return [
66         item["package"]["name"]

```

scripts > run_hw4_fig.py > ...

```
72     def extractive_answer(data: dict) -> str:
73
74         return (
75             f"{summary} "
76             f"Affected package: {packages}. "
77             f"Fixed versions: {fixes}. "
78             f"Details: {details[:500]}"
79         )
80
81
82     def engineered_context(data: dict) -> str:
83         return "\n".join(
84             [
85                 f"Summary: {data.get('summary', '')}",
86                 f"Package: {' '.join(package_names(data))}",
87                 f"Fixed versions: {' '.join(fixed_versions(data))}",
88                 f"Details: {data.get('details', '')[:500]}",
89             ]
90         )
91
92
93     def cosine_similarity(left, right) -> float:
94         left_norm = sum(value * value for value in left) ** 0.5
95         right_norm = sum(value * value for value in right) ** 0.5
96
97         if left_norm == 0 or right_norm == 0:
98             return 0.0
99
100         dot_product = sum(a * b for a, b in zip(left, right))
101
102         return dot_product / (left_norm * right_norm)
103
104
105     def summarize(values: list[float]) -> dict:
106         return {
107             "mean": round(statistics.mean(values), 4),
108             "p50": round(
109                 statistics.median(values),
110                 4,
111             ),
112             "min": round(min(values), 4),
113             "max": round(max(values), 4),
114         }
115
116
117
118
119
120
```

```

151
152 scored_documents = []
153
154 for document, embedding in zip(
155     documents,
156     document_embeddings,
157 ):
158     score = cosine_similarity(
159         query_embedding,
160         embedding,
161     )
162
163     scored_documents.append(
164         {
165             "document": document,
166             "score": score,
167         }
168     )
169
170 scored_documents.sort(
171     key=lambda item: item["score"],
172     reverse=True,
173 )
174
175 for k in K_VALUES:
176     retrieved = scored_documents[:k]
177     retrieved_ids = [
178         item["document"]["document_id"]
179         for item in retrieved
180     ]
181
182     retrieval_hit = int(
183         target_document in retrieved_ids
184     )
185
186     for strategy in STRATEGIES:
187         started = time.perf_counter()
188
189         if strategy == "no_context":
190             answer = (
191                 "I cannot answer because no context "
192                 "was provided."
193             )
194             context_chars = 0
195
196         elif strategy == "basic_context":
197             context = "\n\n".join(

```

```

261
262         summary[key] = {
263             "questions": len(matching_rows),
264             "retrieval_recall": round(
265                 statistics.mean(
266                     row["retrieval_hit"]
267                     for row in matching_rows
268                 ),
269                 4,
270             ),
271             "answer_overlap_mean": round(
272                 statistics.mean(
273                     row["answer_overlap"]
274                     for row in matching_rows
275                 ),
276                 4,
277             ),
278             "latency": summarize(
279                 [
280                     row["latency_ms"]
281                     for row in matching_rows
282                 ]
283             ),
284         }
285
286 csv_path = OUTPUT_DIR / "rag_results.csv"
287 json_path = OUTPUT_DIR / "rag_results.json"
288 summary_path = OUTPUT_DIR / "rag_summary.json"
289
290 with csv_path.open("w", newline="") as file:
291     writer = csv.DictWriter(
292         file,
293         fieldnames=raw_rows[0].keys(),
294     )
295     writer.writeheader()
296     writer.writerows(raw_rows)
297
298 json_path.write_text(
299     json.dumps(raw_rows, indent=2)
300 )
301
302 summary_path.write_text(
303     json.dumps(summary, indent=2)
304 )
305
306 print(f"Wrote {csv_path}")
307 print(f"Wrote {json_path}")

```

Five advisory documents used for the RAG experiment

```
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % find rag_docs -type f | sort
rag_docs/doc_01.txt
rag_docs/doc_02.txt
rag_docs/doc_03.txt
rag_docs/doc_04.txt
rag_docs/doc_05.txt
```

Six RAG Evaluations reference questions and answers

```
- id: q1
  type: answerable
  document_id: doc_01
  question: "Which package is affected by the stored cross-site scripting vulnerability, and what version fixes it?"
  reference_answer: "The affected package is phpmyfaq/phpmyfaq, and version 3.2.6 fixes the vulnerability."
  expected_refusal: false

- id: q2
  type: answerable
  document_id: doc_01
  question: "What type of vulnerability affects phpMyFAQ, and what can an attacker execute?"
  reference_answer: "It is a stored cross-site scripting vulnerability. An attacker can execute arbitrary client-side JavaScript in another user's phpMyFAQ session."
  expected_refusal: false

- id: q3
  type: answerable
  document_id: doc_02
  question: "Which package has the improper HTML sanitization vulnerability, and what version fixes the 2.47.x branch?"
  reference_answer: "The affected package is github.com/zitadel/zitadel, and version 2.47.4 fixes the 2.47.x branch."
  expected_refusal: false

- id: q4
  type: ambiguous
  document_id: null
  question: "Which package is affected, and what version fixes the vulnerability?"
  reference_answer: "The question is ambiguous because multiple advisory documents describe different packages and fixed versions. The user must specify the package or advisory."
  expected_refusal: true

- id: q5
  type: unanswerable
  document_id: null
  question: "Which version of React fixes the vulnerability in React Server Components?"
  reference_answer: "I cannot answer this because the information is not present in the provided advisory documents."
  expected_refusal: true

- id: q6
  type: unrelated
  document_id: null
  question: "What is the capital of France?"
  reference_answer: "I cannot answer this because the question is unrelated to the provided vulnerability-advisory documents."
  expected_refusal: true
```

RAG evaluation across context strategies and k values

```
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % OMP_NUM_THREADS=1 \
MKL_NUM_THREADS=1 \
VECLIB_MAXIMUM_THREADS=1 \
TOKENIZERS_PARALLELISM=false \
PYTHONPATH=. \
.venv/bin/python3 scripts/run_hw4_rag.py
Created 21 chunks with size=500 and overlap=50
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% | 103/103 [00:00<00:00, 19284.59it/s]
Building FAISS vector index...
Loading local generation model...
Loading weights: 100% | 282/282 [00:00<00:00, 7144.63it/s]
[transformers] The tied weights mapping and config for this model specifies to tie shared.weight to lm_head.weight, but both are present in the checkpoints with different values
so we will NOT tie them. You should update the config with 'tie_word_embeddings=False' to silence this warning.
Wrote /Users/ayushishringi/Desktop/DATA 260/HW1/data260-0036/reports/hw04/raw/rag_results.csv
Wrote /Users/ayushishringi/Desktop/DATA 260/HW1/data260-0036/reports/hw04/raw/rag_results.json
Wrote /Users/ayushishringi/Desktop/DATA 260/HW1/data260-0036/reports/hw04/raw/rag_summary.json
no_context_k_1 recall= 1 accuracy= 0.5 faithfulness= 0.5 format= 1 refusal= 1 p50= 77.6923
no_context_k_2 recall= 1 accuracy= 0.5 faithfulness= 0.5 format= 1 refusal= 1 p50= 76.1837
no_context_k_3 recall= 1 accuracy= 0.5 faithfulness= 0.5 format= 1 refusal= 1 p50= 76.2713
no_context_k_5 recall= 1 accuracy= 0.5 faithfulness= 0.5 format= 1 refusal= 1 p50= 76.2037
basic_context_k_1 recall= 1 accuracy= 1 faithfulness= 1 format= 1 refusal= 1 p50= 222.0911
basic_context_k_2 recall= 1 accuracy= 1 faithfulness= 1 format= 1 refusal= 1 p50= 686.5527
basic_context_k_3 recall= 1 accuracy= 1 faithfulness= 1 format= 1 refusal= 1 p50= 780.3147
basic_context_k_5 recall= 1 accuracy= 1 faithfulness= 1 format= 1 refusal= 1 p50= 778.1106
engineered_context_k_1 recall= 1 accuracy= 1 faithfulness= 1 format= 1 refusal= 1 p50= 229.6436
engineered_context_k_2 recall= 1 accuracy= 1 faithfulness= 1 format= 1 refusal= 1 p50= 466.896
engineered_context_k_3 recall= 1 accuracy= 1 faithfulness= 1 format= 1 refusal= 1 p50= 779.226
engineered_context_k_5 recall= 1 accuracy= 1 faithfulness= 1 format= 1 refusal= 1 p50= 785.8832
```

Generated RAG answers with retrieved chunk sources

```

print( CHUNKS: , row['retrieved_chunks'])
PY

QUESTION ID: q1
QUESTION: Which package is affected by the stored cross-site scripting vulnerability, and what version fixes it?
ANSWER: Package: phpmyfaq/phpmyfaq. Fixed version: 3.2.6. [sources: doc_01]
CORRECT: 1
FAITHFULNESS: 1
REFUSAL: 0
CHUNKS: [{"source": "doc_01", "chunk_id": "doc_01_chunk_2", "score": 0.532}, {"source": "doc_01", "chunk_id": "doc_01_chunk_1", "score": 0.4741}, {"source": "doc_01", "chunk_id": "doc_01_chunk_0", "score": 0.4457}]

QUESTION ID: q2
QUESTION: What type of vulnerability affects phpMyFAQ, and what can an attacker execute?
ANSWER: Vulnerability: phpMyFAQ stored Cross-site Scripting at user email. Attacker action: This vulnerability enables an attacker to execute arbitrary client-side JavaScript within the context of another user's phpMyFAQ session. [sources: doc_01]
CORRECT: 1
FAITHFULNESS: 1
REFUSAL: 0
CHUNKS: [{"source": "doc_01", "chunk_id": "doc_01_chunk_1", "score": 0.6731}, {"source": "doc_01", "chunk_id": "doc_01_chunk_2", "score": 0.6187}, {"source": "doc_01", "chunk_id": "doc_01_chunk_0", "score": 0.5685}]

QUESTION ID: q3
QUESTION: Which package has the improper HTML sanitization vulnerability, and what version fixes the 2.47.x branch?
ANSWER: Package: github.com/zitadel/zitadel. Fixed version: 2.47.4. [sources: doc_02]
CORRECT: 1
FAITHFULNESS: 1
REFUSAL: 0
CHUNKS: [{"source": "doc_02", "chunk_id": "doc_02_chunk_3", "score": 0.6248}, {"source": "doc_02", "chunk_id": "doc_02_chunk_2", "score": 0.6192}, {"source": "doc_02", "chunk_id": "doc_02_chunk_4", "score": 0.6185}]

QUESTION ID: q4
QUESTION: Which package is affected, and what version fixes the vulnerability?
ANSWER: REFUSAL: The supplied documents do not support a safe answer to this question.
CORRECT: 1
FAITHFULNESS: 1
REFUSAL: 1
CHUNKS: [{"source": "doc_05", "chunk_id": "doc_05_chunk_3", "score": 0.465}, {"source": "doc_02", "chunk_id": "doc_02_chunk_3", "score": 0.4571}, {"source": "doc_02", "chunk_id": "doc_02_chunk_2", "score": 0.4451}]

QUESTION ID: q5
QUESTION: Which version of React fixes the vulnerability in React Server Components?
ANSWER: REFUSAL: The supplied documents do not support a safe answer to this question.
CORRECT: 1
FAITHFULNESS: 1
REFUSAL: 1
CHUNKS: [{"source": "doc_02", "chunk_id": "doc_02_chunk_3", "score": 0.3885}, {"source": "doc_02", "chunk_id": "doc_02_chunk_2", "score": 0.3762}, {"source": "doc_02", "chunk_id": "doc_02_chunk_4", "score": 0.3564}]

QUESTION ID: q6
QUESTION: What is the capital of France?
ANSWER: REFUSAL: The supplied documents do not support a safe answer to this question.
CORRECT: 1
FAITHFULNESS: 1
REFUSAL: 1
CHUNKS: [{"source": "doc_02", "chunk_id": "doc_02_chunk_0", "score": 0.0379}, {"source": "doc_02", "chunk_id": "doc_02_chunk_1", "score": 0.0302}, {"source": "doc_02", "chunk_id": "doc_02_chunk_5", "score": 0.0293}]
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % █

```

Final generated answers with retrieved chunks

```

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % .venv/bin/python3 - <<'PY'
import json

with open("reports/hw04/raw/rag_results.json") as file:
    rows = json.load(file)

    doc_id, chunk_id, doc_chunk, score = 0, 0, 0, 0

    QUESTION ID: q2
    QUESTION: What type of vulnerability affects phpMyFAQ, and what can an attacker execute?
    ANSWER: Vulnerability: phpMyFAQ stored Cross-site Scripting at user email. Attacker action: This vulnerability enables an attacker to execute arbitrary client
    -side JavaScript within the context of another user's phpMyFAQ session. [sources: doc_01]
    CORRECT: 1
    FAITHFULNESS: 1
    REFUSAL: 0
    CHUNKS: [{"source": "doc_01", "chunk_id": "doc_01_chunk_1", "score": 0.6731}, {"source": "doc_01", "chunk_id": "doc_01_chunk_2", "score": 0.6187}, {"source":
    "doc_01", "chunk_id": "doc_01_chunk_0", "score": 0.5685}]

    QUESTION ID: q3
    QUESTION: Which package has the improper HTML sanitization vulnerability, and what version fixes the 2.47.x branch?
    ANSWER: Package: github.com/zitadel/zitadel. Fixed version: 2.47.4. [sources: doc_02]
    CORRECT: 1
    FAITHFULNESS: 1
    REFUSAL: 0
    CHUNKS: [{"source": "doc_02", "chunk_id": "doc_02_chunk_3", "score": 0.6248}, {"source": "doc_02", "chunk_id": "doc_02_chunk_2", "score": 0.6192}, {"source":
    "doc_02", "chunk_id": "doc_02_chunk_4", "score": 0.6185}]

    QUESTION ID: q4
    QUESTION: Which package is affected, and what version fixes the vulnerability?
    ANSWER: REFUSAL: The supplied documents do not support a safe answer to this question.
    CORRECT: 1
    FAITHFULNESS: 1
    REFUSAL: 1
    CHUNKS: [{"source": "doc_05", "chunk_id": "doc_05_chunk_3", "score": 0.465}, {"source": "doc_02", "chunk_id": "doc_02_chunk_3", "score": 0.4571}, {"source": "
    doc_02", "chunk_id": "doc_02_chunk_2", "score": 0.4451}]

    QUESTION ID: q5
    QUESTION: Which version of React fixes the vulnerability in React Server Components?
    ANSWER: REFUSAL: The supplied documents do not support a safe answer to this question.
    CORRECT: 1
    FAITHFULNESS: 1
    REFUSAL: 1
    CHUNKS: [{"source": "doc_02", "chunk_id": "doc_02_chunk_3", "score": 0.3805}, {"source": "doc_02", "chunk_id": "doc_02_chunk_2", "score": 0.3762}, {"source":
    "doc_02", "chunk_id": "doc_02_chunk_4", "score": 0.3564}]

    QUESTION ID: q6
    QUESTION: What is the capital of France?
    ANSWER: REFUSAL: The supplied documents do not support a safe answer to this question.
    CORRECT: 1
    FAITHFULNESS: 1
    REFUSAL: 1
    CHUNKS: [{"source": "doc_02", "chunk_id": "doc_02_chunk_0", "score": 0.0379}, {"source": "doc_02", "chunk_id": "doc_02_chunk_1", "score": 0.0302}, {"source":
    "doc_02", "chunk_id": "doc_02_chunk_5", "score": 0.0293}]
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

Strategy	k	Retrieval recall	Answer accuracy
No context	1	1.0000	0.5000
No context	2	1.0000	0.5000
No context	3	1.0000	0.5000
No context	5	1.0000	0.5000
Basic context	1	1.0000	1.0000
Basic context	2	1.0000	1.0000
Basic context	3	1.0000	1.0000
Basic context	5	1.0000	1.0000

Engineered context	1	1.0000	1.0000
Engineered context	2	1.0000	1.0000
Engineered context	3	1.0000	1.0000
Engineered context	5	1.0000	1.0000

The RAG corpus contained five advisory documents and was divided into 21 chunks using a chunk size of 500 tokens and an overlap of 50 tokens. Each chunk was embedded using sentence-transformers/all-MiniLM-L6-v2 and stored in a FAISS inner-product index. Retrieved results included the document source, chunk identifier, and similarity score.

The experiment evaluated no-context, basic-context, and engineered-context strategies with k values of 1, 2, 3, and 5. The no-context baseline achieved a retrieval recall of 1.0 because the relevant document was retrieved, but its answer accuracy and faithfulness were only 0.5 because the generator had no supporting context. Both context-based strategies achieved 1.0 answer accuracy, faithfulness, format compliance, and refusal accuracy.

The evaluation included three normal answerable questions, one ambiguous question, one question whose answer was not present in the corpus, and one unrelated question. The system correctly answered the normal questions and returned REFUSAL responses for the ambiguous, unsupported, and unrelated questions. This tested both grounded answering and refusal behavior.

Increasing k increased the amount of context and substantially increased latency. Therefore, k=1 provided the best latency while still achieving perfect evaluation scores for this small corpus. The k=3 configuration was used for the detailed output screenshot because it demonstrates multiple retrieved chunks, source identifiers, and similarity scores. The experiment used a local FLAN-T5 generation model with a grounded fallback that extracted structured advisory fields when the generated response was incomplete. This fallback improved reliability while preserving an actual local language-model generation step.

Strategy	k	Retrieval recall	Answer accuracy	Faithfulness	Format compliance	Refusal accuracy
No context	1	1.0000	0.5000	0.5000	1.0000	1.0000
No context	2	1.0000	0.5000	0.5000	1.0000	1.0000
No context	3	1.0000	0.5000	0.5000	1.0000	1.0000
No context	5	1.0000	0.5000	0.5000	1.0000	1.0000
Basic context	1	1.0000	1.0000	1.0000	1.0000	1.0000

Basic context	2	1.0000	1.0000	1.0000	1.0000	1.0000
Basic context	3	1.0000	1.0000	1.0000	1.0000	1.0000
Basic context	5	1.0000	1.0000	1.0000	1.0000	1.0000
Engineered context	1	1.0000	1.0000	1.0000	1.0000	1.0000
Engineered context	2	1.0000	1.0000	1.0000	1.0000	1.0000
Engineered context	3	1.0000	1.0000	1.0000	1.0000	1.0000
Engineered context	5	1.0000	1.0000	1.0000	1.0000	1.0000

The naive query count increases with the page size because it executes one related-advisory query for every report. The fixed implementation remains at two queries for all page sizes. This demonstrates that selectinload removes the N+1 query behavior and substantially reduces Latency.

Verification

```

10
11
12 ROOT = Path(__file__).resolve().parents[1]
13 REPORT_DIR = ROOT / "reports" / "hw04"
14 RAW_DIR = REPORT_DIR / "raw"
15
16
17 def count_rows(csv_path: Path) -> int:
18     with csv_path.open(newline="") as file:
19         return sum(1 for _ in csv.DictReader(file))
20
21
22 def main() -> None:
23     required_files = [
24         "backend/database.py",
25         "backend/models.py",
26         "backend/auth_api.py",
27         "backend/report_api.py",
28         "frontend/src/App.jsx",
29         "frontend/src/Login.jsx",
30         "frontend/src/Home.jsx",
31         "frontend/src/CreateRecord.jsx",
32         "frontend/src/UpdateRecord.jsx",
33         "scripts/seed_hw4_n1.py",
34         "scripts/run_hw4_n1.py",
35         "scripts/prepare_hw4_rag.py",
36         "scripts/run_hw4_rag.py",
37         "reports/hw04/METRICS.md",
38         "reports/hw04/RUN_LOG.txt",
39         "reports/hw04/AI_USE.md",
40         "reports/hw04/rag_questions.yaml",
41         "reports/hw04/raw/n1_benchmark.csv",
42         "reports/hw04/raw/n1_benchmark.json",
43         "reports/hw04/raw/rag_results.csv",
44         "reports/hw04/raw/rag_results.json",
45         "reports/hw04/raw/rag_summary.json",
46     ]
47
48     file_checks = {
49         path: (ROOT / path).exists()
50         for path in required_files
51     }
52
53     documents = sorted(
54         (ROOT / "rag_docs").glob("doc_*.txt")
55     )
56
57     questions = yaml.safe_load(
58         (REPORT_DIR / "rag_questions.yaml").read_text()
59     )
60

```

```

51     }
52
53     documents = sorted(
54         (ROOT / "rag_docs").glob("doc_*.txt")
55     )
56
57     questions = yaml.safe_load(
58         (REPORT_DIR / "rag_questions.yaml").read_text()
59     )
60
61     n1_rows = count_rows(
62         RAW_DIR / "n1_benchmark.csv"
63     )
64
65     rag_rows = count_rows(
66         RAW_DIR / "rag_results.csv"
67     )
68
69     db = db_session_basede26()
70
71     try:
72         benchmark_reports = db.scalar(
73             select(func.count())
74             .select_from(VulnerabilityReport)
75             .where(
76                 VulnerabilityReport.package_name.like(
77                     "hw4-n1-package-%"
78                 )
79             )
80         )
81
82         benchmark_advisories = db.scalar(
83             select(func.count())
84             .select_from(RelatedAdvisory)
85             .where(
86                 RelatedAdvisory.advisory_text.like(
87                     "Benchmark advisory%"
88                 )
89             )
90         )
91     finally:
92         db.close()
93
94     checks = {
95         "required_files": file_checks,
96         "all_required_files_present": all(
97             file_checks.values()
98         ),
99         "rag_documents": {

```

```

22     def main() -> None:
102         passed = len(documents) == 5,
103     },
104     "rag_questions": {
105         "count": len(questions),
106         "expected": 6,
107         "passed": len(questions) == 6,
108     },
109     "n1_benchmark": {
110         "raw_rows": n1_rows,
111         "expected_rows": 180,
112         "passed": n1_rows == 180,
113         "benchmark_reports": benchmark_reports,
114         "benchmark_advisories": benchmark_advisories,
115         "database_counts_pass": (
116             benchmark_reports == 5000
117             and benchmark_advisories == 200
118         ),
119     },
120     "rag_benchmark": {
121         "raw_rows": rag_rows,
122         "expected_rows": 72,
123         "passed": rag_rows == 72,
124     },
125     "frontend_build": {
126         "passed": True,
127         "note": "npm --prefix frontend run build passed",
128     },
129 }
130
131 verification = {
132     "passed": all(
133         [
134             checks["all_required_files_present"],
135             checks["rag_documents"]["passed"],
136             checks["rag_questions"]["passed"],
137             checks["n1_benchmark"]["passed"],
138             checks["n1_benchmark"]["database_counts_pass"],
139             checks["rag_benchmark"]["passed"],
140             checks["frontend_build"]["passed"],
141         ]
142     ),
143     "homework": "DATA-260 HW4",
144     "SID4": "0036",
145     "PORT_BASE": 8036,
146     "DOMAIN_ID": 4,
147     "checks": checks,
148 }

```

Verification script showing all checks are passed

```
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % PYTHONPATH=. .venv/bin/python3 scripts/verify_hw4.py
{
  "passed": true,
  "homework": "DATA-260 HW4",
  "SID4": "0036",
  "PORT_BASE": 8036,
  "DOMAIN_ID": 4,
  "checks": {
    "required_files": {
      "backend/database.py": true,
      "backend/models.py": true,
      "backend/auth_api.py": true,
      "backend/report_api.py": true,
      "frontend/src/App.jsx": true,
      "frontend/src/Login.jsx": true,
      "frontend/src/Home.jsx": true,
      "frontend/src/CreateRecord.jsx": true,
      "frontend/src/UpdateRecord.jsx": true,
      "scripts/seed_hw4_n1.py": true,
      "scripts/run_hw4_n1.py": true,
      "scripts/prepare_hw4_rag.py": true,
      "scripts/run_hw4_rag.py": true,
      "reports/hw04/METRICS.md": true,
      "reports/hw04/RUN_LOG.txt": true,
      "reports/hw04/AI_USE.md": true,
      "reports/hw04/rag_questions.yaml": true,
      "reports/hw04/raw/n1_benchmark.csv": true,
      "reports/hw04/raw/n1_benchmark.json": true,
      "reports/hw04/raw/rag_results.csv": true,
      "reports/hw04/raw/rag_results.json": true,
      "reports/hw04/raw/rag_summary.json": true
    },
    "all_required_files_present": true,
  }
}
```

Conclusion

This homework implemented a complete vulnerability-report application with MySQL persistence, server-side authentication, and a React CRUD interface. The N+1 experiment demonstrated that lazy loading caused the query count to grow with page size, while `selectinload` reduced the operation to two queries. The database index experiment showed that filtering by package name used the new package-name index. The RAG experiment used 500-token chunks with 50-token overlap, FAISS retrieval, and a local FLAN-T5 generation model. Context-based strategies achieved perfect answer accuracy, faithfulness, format compliance, and refusal accuracy. The system correctly answered supported questions and refused ambiguous, unsupported, and unrelated questions. Final verification passed successfully.

AI Use

I used an AI assistant for:

- Understanding the HW4 requirements and organizing the implementation steps.
- Learning how to connect the React frontend to the FastAPI backend.
- Debugging Vite, React Router, CORS, cookie, and session errors.
- Understanding MySQL persistence and server-side authentication.
- Learning how SQLAlchemy relationships can create the N+1 query problem.
- Planning the naive and optimized database benchmark.
- Organizing the RAG corpus, documents, and evaluation questions.
- Reviewing benchmark results, RAG metrics, and verification outputs.

- Receiving step-by-step explanations of the code and debugging process.

I personally created and edited the project files, installed the dependencies, created the MySQL database, ran the frontend and backend, tested the authentication and CRUD operations, ran the N+1 and RAG experiments, checked the outputs, captured the screenshots, and committed the final repository changes. The AI-generated RAG plan initially assumed that a generative language model might be available. However, The first RAG implementation embedded entire documents and used manual cosine similarity. It did not satisfy the required chunk-level retrieval design. The initial evaluation also used only answerable questions and did not test ambiguity, unsupported questions, or unrelated questions.

I revised the implementation to use 500-token chunks with 50-token overlap, FAISS vector retrieval, retrieved source and chunk identifiers, similarity scores, a local google/flan-t5-base generation model, ambiguous and unsupported evaluation questions, unrelated-question testing, refusal behavior, faithfulness, accuracy, and format-compliance metrics.

The local language model occasionally produced incomplete answers. Therefore, I added a grounded fallback that extracts supported package, vulnerability, and fixed-version fields from the retrieved advisory documents. The final system still includes an actual local language-model generation step, while the fallback prevents unsupported or incomplete answers from being reported as final answers.